

CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

Project Documentation

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Tools Used: Python, Google Colab, Power BI, GitHub

Introduction

Customer churn refers to the percentage of customers who stop using a company's products or services within a specific period. In today's competitive business environment, customer retention is one of the most important factors for organizational success.

Acquiring new customers is generally more expensive than retaining existing ones, making churn prediction a valuable business strategy.

The banking industry handles millions of customer accounts, making it difficult to manually identify customers who are likely to leave. By using Machine Learning techniques, organizations can analyze historical customer data and predict churn in advance. This allows businesses to take proactive actions such as offering personalized services, discounts, or loyalty programs to retain valuable customers.

The **Customer Churn Prediction** project applies supervised machine learning algorithms to analyze customer information and predict whether a customer is likely to leave the bank. The project also includes an interactive Power BI dashboard that helps stakeholders visualize customer behavior and churn trends.

Project Objective

The primary objective of this project is to develop a Machine Learning model capable of accurately predicting customer churn based on historical customer information.

The specific objectives include:

- Understanding customer behavior through data analysis.
- Cleaning and preprocessing raw customer data.
- Performing Exploratory Data Analysis (EDA) to discover hidden patterns.
- Identifying the most influential features affecting customer churn.
- Building and comparing multiple Machine Learning classification models.
- Evaluating model performance using standard evaluation metrics.
- Selecting the best-performing model for prediction.
- Creating an interactive Power BI dashboard for business decision-making.

Problem Statement

Customer churn is one of the major challenges faced by banks and financial institutions. Every year, organizations lose a significant number of customers due to various reasons such as poor customer experience, competitive offers, high service charges, or lack of customer engagement.

Since manually identifying customers who may leave is difficult, organizations require an intelligent prediction system that can automatically analyze customer data and estimate the probability of churn.

This project addresses this problem by developing a Machine Learning-based predictive model that classifies customers into two categories:

- Customers who are likely to stay.
- Customers who are likely to leave.

The predicted information can help organizations improve customer retention strategies and reduce revenue loss.

Dataset Description

The project uses the **Churn_Modelling.csv** dataset, which contains historical customer information collected from a banking institution.

Dataset Overview

Parameter	Value
Dataset Name	Churn_Modelling.csv
Total Records	10,000
Total Columns	14
Target Variable	Exited

Dataset Features

Feature	Description
Credit Score	Customer's credit score
Geography	Customer's country
Gender	Male or Female
Age	Customer age
Tenure	Years with the bank
Balance	Account balance

Num Of Products

Has Cr Card

Is Active Member

Estimated Salary

Exited

Number of products used

Credit card status

Customer activity status

Annual estimated salary

Customer churn status

Target Variable:

- **0** → Customer Stayed
- **1** → Customer Left

Tools and Technologies Used

The project was developed using the following tools and technologies:

Programming Language

- Python

Development Environment

- Google Colab

Data Analysis Libraries

- Pandas
- NumPy

Data Visualization Libraries

- Matplotlib
- Seaborn

Machine Learning Library

- Scikit-learn

Model Saving

- Joblib

Dashboard Development

- Power BI Desktop

Methodology

The project follows a complete Machine Learning workflow from data collection to prediction and visualization.

Step 1: Import Required Libraries

All necessary Python libraries were imported, including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. These libraries provide functionalities for data manipulation, visualization, model training, and evaluation.

Step 2: Data Collection

The customer churn dataset was imported into Google Colab using Pandas. Initial inspection of the dataset was performed to understand its structure, number of records, columns, and data types.

Step 3: Data Understanding

The dataset was explored using:

- Head()
- Shape
- Columns
- Info()
- Describe()

This step helped understand:

- Number of rows and columns
- Numerical and categorical features
- Missing values
- Statistical summary

Step 4: Data Cleaning

The raw dataset contained unnecessary columns such as Row Number, Customer Id, and Surname. These columns were removed because they do not contribute to churn prediction.

The dataset was checked for:

- Missing values
- Duplicate records
- Incorrect data types

After cleaning, the dataset became suitable for Machine Learning.

Step 5: Exploratory Data Analysis (EDA)

EDA was performed to understand customer behavior and identify patterns related to churn.

Several visualizations were created, including:

- Customer Churn Distribution
- Gender-wise Churn Analysis
- Geography-wise Churn Analysis
- Age Distribution
- Credit Score Distribution
- Balance Distribution
- Active Member Analysis
- Correlation Heatmap

The analysis revealed that customer age, balance, geography, and active membership significantly influence churn.

Step 6: Data Preprocessing

Before training the model, the dataset underwent preprocessing.

The preprocessing included:

- Label Encoding categorical variables
- Feature Selection
- Train-Test Split (80:20)
- Feature Scaling using Standard Scaler

These steps improve model performance and prediction accuracy.

Step 7: Machine Learning Model Development

Three classification algorithms were implemented.

Logistic Regression

A linear classification algorithm used as the baseline model.

Decision Tree Classifier

Used to capture complex nonlinear relationships between customer features.

Random Forest Classifier

An ensemble learning algorithm that combines multiple decision trees to achieve higher accuracy and reduce overfitting.

Step 8: Model Evaluation

The trained models were evaluated using multiple performance metrics.

Evaluation Metrics:

- Accuracy Score
- Confusion Matrix
- Precision
- Recall
- F1-Score
- Classification Report
- ROC-AUC Score
- ROC Curve

These metrics help determine the effectiveness of each model.

Step 9: Feature Importance Analysis

Feature Importance analysis was performed using the Random Forest model.

The most influential features included:

- Age
- Balance
- Number of Products
- Geography
- Credit Score
- Active Member Status
- Estimated Salary

This analysis helps understand which customer characteristics contribute most to churn.

Step 10: Model Comparison

All Machine Learning models were compared based on their prediction accuracy and evaluation metrics.

Among all implemented models, the **Random Forest Classifier** achieved the highest accuracy and provided the most reliable predictions.

Therefore, Random Forest was selected as the final prediction model.

Step 11: Model Saving

The trained Random Forest model was saved using the Joblib library.

Saving the model allows it to be reused for future predictions without retraining.

Power BI Dashboard

A professional and interactive Power BI dashboard was created using the cleaned customer dataset.

The dashboard provides business insights through interactive visualizations.

KPI Cards

- Total Customers
- Churn Customers
- Retained Customers
- Churn Rate
- Average Age
- Average Balance

Visualizations

- Churn by Gender
- Churn by Geography
- Churn by Number of Products
- Age vs Churn

- Balance vs Churn
- Credit Score Distribution
- Active Members vs Churn

Interactive Filters

- Geography
- Gender
- Active Member
- Number of Products
- Churn Status

The dashboard enables management to monitor churn trends and make informed decisions.

Results

The project successfully achieved the following outcomes:

- Cleaned and prepared the customer dataset.
- Performed detailed Exploratory Data Analysis.
- Built three Machine Learning models.
- Compared model performance using multiple evaluation metrics.
- Identified Random Forest as the best-performing algorithm.
- Saved the trained prediction model.
- Developed a professional Power BI dashboard for business reporting.

Advantages

The developed system provides several advantages:

- Accurate customer churn prediction.
- Early identification of at-risk customers.
- Supports data-driven business decisions.
- Improves customer retention strategies.
- Reduces business revenue loss.
- Interactive dashboard for quick analysis.
- Scalable for large customer datasets.

Applications

Customer churn prediction can be applied in various industries, including:

- Banking
- Insurance
- Telecommunications
- E-commerce
- Retail
- Subscription-based businesses
- Healthcare
- Online service platforms

Future Scope

The project can be further enhanced by:

- Deploying the model using Streamlit or Flask.
- Integrating with real-time customer databases.
- Implementing advanced algorithms such as XGBoost and LightGBM.
- Developing automated customer retention recommendation systems.
- Creating cloud-based prediction services.
- Integrating with CRM software for real-time monitoring.

Conclusion

The **Customer Churn Prediction using Machine Learning** project demonstrates the practical application of data analytics and machine learning in solving real-world business problems. By analyzing customer demographics, financial information, and account activity, the project successfully predicts customers who are likely to leave the bank. Data preprocessing, exploratory analysis, feature engineering, and model evaluation ensured reliable predictions, while the Random Forest Classifier emerged as the most effective model due to its high accuracy and robustness. The interactive Power BI dashboard further enhances the project by providing meaningful visual insights into customer behavior and churn trends. Overall, this project enables organizations to make proactive business

decisions, improve customer satisfaction, reduce churn, and increase long-term profitability.